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“Bidirectional Mass Spectrometry Modeling Aided by Auxiliary Intermediate Steps From Graph Transformation Tools”

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Abstract

Mass spectrometry is a cornerstone technique in analytical chemistry, metabolomics, pharmaceutical research and many other areas. The ability to predict a molecular structure from a given spectrum, and vice versa, remains a central challenge. While forward prediction of spectra from known structures is increasingly accurate due to machine learning methods, the inverse problem — predicting possible molecular structures from a given spectrum — remains difficult, ambiguous, and poorly interpretable.

The work proposes the integration of aiding intermediate fragments, which are generated by graph transformation tools, as a strategy to enhance machine learning-based bidirectional translation between molecules and spectra. The aim is to enhance model interpretability and accuracy.

The work will offer a systematic method for integrating graph-based intermediates into machine learning pipelines, potentially improving interpretability and performance of structure $\leftrightarrow$ spectrum translation models.

Zusammenfassung

Massenspektrometrie ist eine grundlegende Technik in der Analytischen Chemie, der Metabolomik, der pharmazeutischen Forschung und in vielen weiteren Bereichen. Die Vorhersage einer Molekülstruktur aus einem gegebenen Spektrum — und umgekehrt — bleibt eine zentrale Herausforderung. Während die Vorhersage von Spektren aus bekannten Strukturen dank Methoden des maschinellen Lernens zunehmend präzise gelingt, ist das inverse Problem — also die Vorhersage möglicher Molekülstrukturen aus einem gegebenen Spektrum — weiterhin schwierig, mehrdeutig und nur schwer interpretierbar.

Die Arbeit schlägt vor, unterstützende Zwischenfragmente, die durch graphbasierte Transformationstools erzeugt werden, als Strategie zu integrieren, um das maschinelle Lernen für die bidirektionale Übersetzung zwischen Molekülen und Spektren zu verbessern. Ziel ist es, die Interpretierbarkeit und Genauigkeit der Modelle zu steigern.

Die Arbeit wird eine systematische Methode zur Integration graphbasierter Zwischenstufen in Machine-Learning-Pipelines bieten, um potenziell die Interpretierbarkeit und Leistungsfähigkeit von Struktur $\leftrightarrow$ Spektrum-Übersetzungsmodellen zu verbessern.

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Chapter 1

Introduction

1.1 Motivation

Mass spectrometry (MS) is one of the most widely used analytical techniques for molecular characterization across chemistry, metabolomics, environmental sciences, and pharmaceutical research. [1, p. 1] Its high sensitivity [2, p. 1764], reproducibility [2, p. 1764], and scalability [2, p. 1764] have enabled large-scale molecular profiling in both research and industrial settings. Modern instrumentation such as Orbitrap, Q-TOF, and triple quadrupole mass spectrometers generates high-throughput spectral data that can be compared against large repositories of molecular fingerprints such as the NIST Mass Spectral Library to support compound identification and profiling [3, pp. 7274-7275].

Despite these advances in data acquisition, the interpretation of MS data remains a significant bottleneck [4, p. 1]. Structural elucidation from spectra is still largely expert-driven, relying on chemical intuition and established fragmentation rules [4, p. 11]. While thousands of spectra can be recorded within a short time frame [4, p. 1], the corresponding structural annotation process remains comparatively slow, manual, and difficult to scale [4, p. 3]. This imbalance creates a fundamental asymmetry between data generation and data analysis.

The intrinsic complexity of molecular fragmentation contributes substantially to this challenge [1, p. 240]. Fragmentation processes are governed by chemically valid bond cleavages and rearrangements [1, pp. 30-31], leading to multiple competing fragmentation pathways. A single molecule may therefore generate numerous fragments through direct cleavages and rearrangement reactions [1, p. 240-241]. Because fragmentation pathways depend sensitively on molecular structure and internal energy [1, p. 26, pp. 30-31], structurally distinct molecules may produce partially overlapping spectral signatures, while small structural variations can induce significant

spectral differences. Consequently, the inverse problem — inferring molecular structure from spectral data — is inherently ill-posed [1, p. 240].

Recent machine learning approaches aim to address this challenge by learning direct mappings between molecular representations and spectral outputs. Graph neural networks [5], sequence-based encoders [6, p. 1573], and generative models have demonstrated promising performance in forward (molecule-to-spectrum) and inverse (spectrum-to-molecule) prediction tasks. However, purely base-data-driven approaches exhibit several limitations:

- They often lack mechanistic interpretability. [7, p. 3& 6]
- They do not explicitly model fragmentation pathways. [7, p. 3]
- They rely on statistical correlations rather than chemically structured priors. [8, p. 966]
- Their generalization to rare or out-of-distribution molecular scaffolds remains limited. [9, p. 12]

In particular, models that treat molecules as flat graphs or string encodings and spectra as unordered peak lists ignore the structured chemical processes underlying fragmentation. This omission reduces interpretability and may hinder robust extrapolation beyond the training distribution. [10, p. 3420]

Fragmentation in mass spectrometry is not an arbitrary stochastic process; it follows chemically constrained transformation sequences. [11, ch. 4& 8] Therefore, incorporating structured representations of fragmentation—such as derivation graphs encoding valid transformation rules—offers a principled way to introduce chemical inductive bias into machine learning models. By explicitly modeling fragment lineage and transformation pathways, it becomes possible to align molecular substructures with spectral evidence in a mechanistically meaningful manner.

This thesis is motivated by the hypothesis that integrating chemically structured derivation graphs into a machine learning framework can improve predictive robustness, interpretability, and bidirectional structure-spectrum modeling. By combining fragment-aware representations with data-driven learning, the aim is to bridge the gap between high-throughput spectral acquisition and scalable, chemically grounded interpretation.

1.2 Objectives

The objective of this thesis is to design, implement, and systematically evaluate a fragment-augmented machine learning framework for improving the bidirectional translation between molecular structures and mass spectra. Specifically, the work investigates how the integration of

graph-based reaction and fragment derivation trees, generated through chemically valid graph transformation rules, can serve as structured and interpretable intermediate representations that enhance both predictive performance and model transparency.

Mass spectrometry-based structure elucidation represents an inherently ill-posed inverse problem: structurally distinct molecules may yield partially overlapping spectral signatures, while minor structural modifications can induce substantial spectral changes. This thesis therefore introduces fragment-level intermediates derived from rule-based graph transformations to inject chemically grounded structure into the learning process.

Instead of directly learning a mapping between molecular graphs and spectral representations, the proposed framework decomposes the task into a structured pipeline in which molecular graphs are transformed into chemically plausible fragment sets, which in turn mediate the relationship to spectral outputs. These fragment sets are embedded into a shared latent space together with molecular and spectral representations, enabling aligned embeddings across the domains.

Ultimately, this thesis aims to demonstrate that incorporating chemically valid graph-based fragment derivations into neural learning pipelines, enhancing both the predictive power and the interpretability of bidirectional structure-spectrum modeling in mass spectrometry and enabling de novo molecular generation from spectral data.

1.3 Scope and Limitations

This thesis is restricted to electron ionization (EI) mass spectrometry, as EI fragmentation reactions are comparatively well understood and mechanistically characterized. [11, ch. 4& 8] This allows the use of chemically informed rule-based graph transformations to generate fragment derivation trees under consistent assumptions. Other ionization techniques are excluded due to their differing fragmentation behavior and the additional modeling complexity they would introduce.

A major limitation arises from the computational cost of graph-based fragment generation. Rule-based graph transformations lead to combinatorial growth in possible fragments, [12, p. 4] especially when multiple reaction rules and recursive fragmentation steps are considered. Ensuring chemical validity and avoiding duplicate graph states further increases computational demands. [13, pp. 2-3]

Due to this scalability constraint, the experimental evaluation is limited to order of magnitude of 100 compounds. While sufficient for demonstrating methodological feasibility and analyzing interpretability, this restricted dataset limits chemical diversity and does not allow

for generalization. Consequently, the results should be interpreted as proof-of-concept validation rather than evidence of large-scale applicability.

Also to be noted is that the reactions are in terms of graph symbols ambiguously defined, and the transformation rules are not exhaustive of all possible fragmentation pathways. This means that the generated fragment derivation trees may not capture all relevant chemical transformations, and the model’s performance may be sensitive to the specific choice of transformation rules. The interpretability analysis is therefore constrained by the assumptions embedded in the rule set, and may not generalize to fragmentation processes that deviate from these assumptions.

Chapter 2

Background and Related Work

2.1 Fundamentals and Capabilities of Mass Spectrometry

Mass spectrometry is a core analytical technique that characterizes chemical species by converting them into gas-phase ions and measuring their mass-to-charge ratio (m/z). [14, p. 5] A mass spectrometer therefore acts as a highly sensitive "molecular balance" whose output – a mass spectrum – reports which ionic species are present and with what relative abundances.

Conceptually, an MS experiment comprises three main steps:

- (i) **ionization**, in which neutral molecules are transferred into the gas phase and charged; [14, p. 1]
- (ii) **mass analysis**, in which ions are separated according to m/z ; and
- (iii) **detection**, in which ion signals are recorded and assembled into a spectrum. [14, p. 5]

Because many compounds form multiple ion species (e.g., adducts [14, p. 21], isotopologues [1, p. 67], fragments), the measured spectrum is not a direct image of the neutral molecule but a structured measurement that must be interpreted in the context of the ionization and fragmentation processes used. [14, p. 16-18]

A key capability of MS is the *identification of unknown compounds*. Even without prior knowledge of the analyte, MS provides constraints that narrow the space of possible identities. Accurate m/z measurements can restrict candidate molecular formulas; isotope patterns (e.g., from ^{13}C , ^{37}Cl , ^{81}Br) further refine these candidates by providing diagnostic abundance ratios and characteristic spacings. [1, p. 263] When fragmentation data are available – either through tandem MS (MS/MS) or through ionization methods that induce fragmentation – structural hypotheses can be tested by matching observed fragment ions and neutral losses to expected

substructures. [1, p. 225] In practice, unknown identification often combines these orthogonal cues with computational searches in spectral libraries and structure databases, where a measured spectrum is matched against reference spectra or predicted fragmentation patterns. [15, p. 8] For complex samples, chromatographic coupling (GC-MS or LC-MS) adds retention time as an additional dimension, [1, p. 475] improving selectivity and reducing confounding effects from co-eluting compounds. [14, p. 233-237]

MS is equally important for the *quantification of known substances*, where the goal is to estimate analyte concentration rather than molecular identity. [14, p. 260] Quantitative MS is grounded in the relationship between the number of detected ions and the amount of analyte introduced into the instrument. [14, p. 261 & 265-266] In idealized settings, ion signal intensity scales proportionally with concentration; in real measurements, this proportionality is skewed. Consequently, reliable quantification typically relies on calibration strategies. *External calibration* constructs a response curve from standards of known concentration, while *internal standardization* – often using isotopically labeled analogs – compensates for variability during sample preparation, injection, and ionization. [14, p. 264-268] The outcome is a method that can measure trace concentrations across wide dynamic ranges, which is central in many applications.

Beyond identity and amount, MS provides a systematic route to *deduce structural features of molecules*. Structural inference is primarily enabled by fragmentation behavior: when a precursor ion breaks apart, the resulting fragment ions reflect chemically plausible bond cleavages and rearrangements, thereby encoding information about functional groups, substructures, and connectivity. [16, p. i49] Certain fragmentation pathways are characteristic – for example, losses of small neutrals such as H₂O, NH₃, or CO can indicate the presence of hydroxyl, amino, or carbonyl functionalities, while specific fragment series can suggest alkyl chain lengths or aromatic substitution patterns. [1, p. 504] Importantly, structural elucidation from MS data is rarely strait forward; rather, it is an inference problem in which multiple candidate structures may explain the same spectral evidence, especially for isomers. [17, p. 53]

As a result, modern workflows increasingly integrate MS measurements with computational models that generate and rank candidate structures based on spectral consistency.

2.2 Principles of Mass Spectrometry

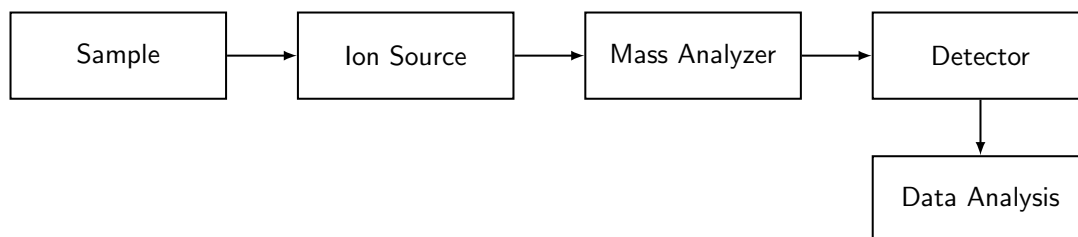


Figure 2.1: Flowchart of a mass spectrometer

[18]

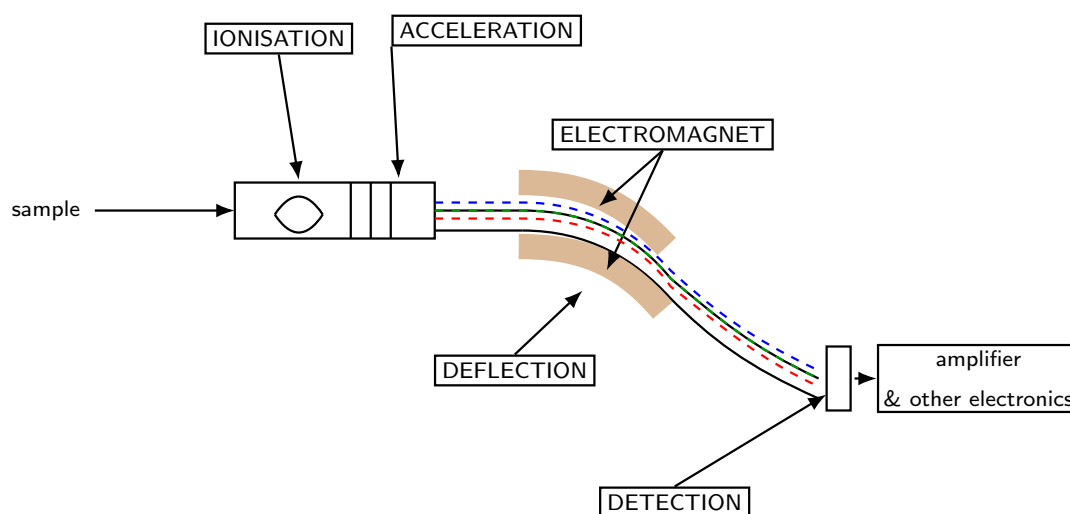


Figure 2.2: Schematic of a mass spectrometer

[19]

Instrument workflow

Figure 2.1 shows steps in a mass spectrometer: a *sample* is introduced, converted into gas-phase ions in the ion source, separated according to their mass-dependent motion in the *mass analyzer*, detected in the *detector*, and finally interpreted during *data analysis*. The central observable is the mass-to-charge ratio m/z . A mass spectrum is therefore a distribution of signal intensity as a function of m/z , where the signal reflects the abundance of ions reaching the detector. [20, p. 1566] In most practical contexts, intensities are reported as relative intensity, normalized such that the most intense peak equals 100% of the base peak. [20, p. 1550] The x-axis is conventionally labeled with m/z (mass-to-charge ratio), a dimensionless quantity. Labeling with mass units such as *daltons* (Da), amu, or unified atomic mass unit (u) is discouraged – although common in practice – as it causes confusion with multiple charged ions. [20, p. 1516]

Ion notation and precursor concepts

Ions observed in mass spectrometry are typically derived from a neutrally charged molecule M . In electron ionization (EI) a hard ionization method, removal of an electron yields the molecular ion [20, p. 1541] – also called the parent ion – denoted $[M^+]$, [20, p. 1582] a radical cation carrying both a positive charge and an unpaired electron. More generally, the term *precursor ion* refers to the ion selected for fragmentation, [20, p. 1578] from which product ions are generated. In soft ionization methods like electrospray, ion formation is often dominated by proton transfer, producing a protonated molecule $[M + H]^+$. [20, p. 1580]

Peak types: isotope, fragment, and base peaks

A measured spectrum contains several conceptually distinct peak types. *Isotope peaks* arise from naturally occurring heavier isotopes (e.g., ^{13}C , ^{37}Cl), producing characteristic $M + 1$, $M + 2$, ... patterns adjacent to the monoisotopic ion peak. *Fragment peaks* arise when precursor ions undergo dissociation, yielding smaller product ions whose m/z values and relative intensities encode structural information. The *base peak* is the most intense peak in the spectrum [20, p. 1524] and is used for normalization; [20, p. 1550] it is not necessarily the molecular ion and often corresponds to a particularly stable fragment. [21, p. 2] [1, p. 263]

2.2.1 Ionization and Fragmentation

Ionization techniques are broadly classified into two categories based on the energy imparted to the analyte: *hard ionization* methods, which deliver high internal energy and promote extensive fragmentation, [20, p. 1547] and *soft ionization* methods, which minimize fragmentation and favor molecular ion preservation. [20, p. 1591] [22, p. 34]

Hard Ionization methods

Electron Ionization (EI) is a hard ionization technique in which a stream of high-energy electrons (typically 70 eV) collides with neutral molecules in the gas phase. [1, p. 197] The collision transfers energy to the molecule, causing the ejection of an electron and formation of a radical cation $[M^+]$ with both a positive charge and an unpaired electron. [1, p. 194] Because the internal energy of the molecular ion typically exceeds the activation barriers for bond cleavage, the $[M^+]$ ion spontaneously fragments into smaller product ions and neutral molecules. [23, p. 3] As a result, EI mass spectra are characterized by rich fragmentation patterns that reflect the molecular structure. [24]

The deterministic nature of EI fragmentation pathways makes electron ionization particularly valuable for structural elucidation. [25] Characteristic losses of small neutral species provide diagnostic clues about functional groups and connectivity. [26] EI is the dominant ionization method in gas chromatography-mass spectrometry (GC-MS) and remains widely used in small-molecule analysis. [27, p. 1]

Spark Ionization (SI) , is a hard ionization technique [28, p. 1], that often utilized to detect trace elements at parts-per-billion (ppb) levels. [29, p. 428] It uses high-voltage up to 100 kV pulsed electric sparks between two electrodes to vaporize, atomize, and ionize conductive or semi-conductive materials. [29, p. 428]

Soft Ionization methods

Electrospray Ionization (ESI) is a prominent soft ionization technique particularly well-suited for coupling with liquid chromatography (LC-MS) and for analysis of large biomolecules. [30] In ESI, a liquid sample is converted into a fine aerosol of charged droplets by applying a high electric potential (typically several kilovolts) between a capillary needle and a counter electrode. [31, p. 3] As the droplets travel toward the mass analyzer, they undergo solvent evaporation and coulombic fission, eventually releasing gas-phase ions into the vacuum of the spectrometer. [31, p. 7] [32, p. 53]

ESI ionization is exceptionally gentle, with minimal or no fragmentation in the ion source. [31, p. 7] This characteristic makes ESI ideal for determining accurate molecular weights of intact molecules. [32, p. 211] Notably, ESI can generate multiply charged ions $[M + nH]^{n+}$ from large molecules, reducing their effective m/z and enabling their analysis on instruments with limited mass range. [32, p. 50]

A curtain gas is often employed to shield the ion path, preventing neutral contaminants and solvent droplets from reaching the mass analyzer while allowing ions to pass through. [14, p. 42] The sequence from charged droplet formation to gas-phase ion release is shown in Figure 2.3.

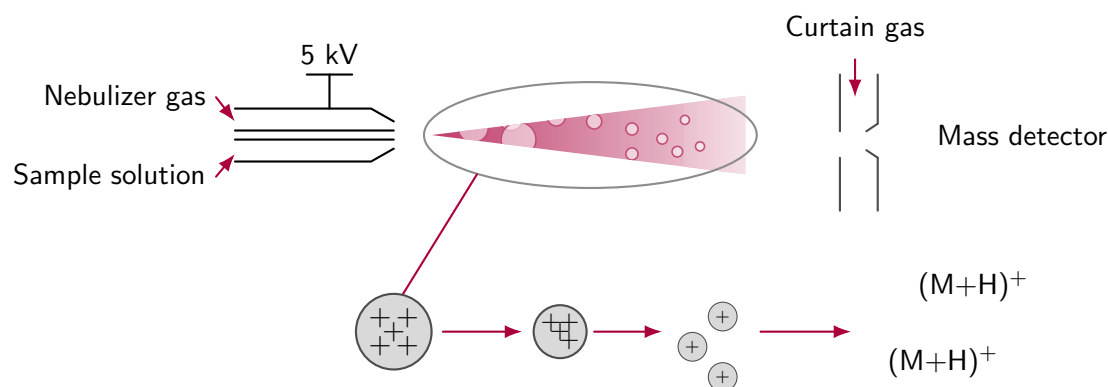


Figure 2.3: Schematic of the electrospray ionization process

[22, p.39]

Atmospheric Pressure Chemical Ionization (APCI) is a soft ionization method used primarily in LC-MS, [22, p. 39] where analytes are ionized at atmospheric pressure via ion-molecule chemistry initiated by a corona discharge. [20, p. 1522] In contrast to ESI, no high voltage is applied to the inlet capillary; instead, the mobile phase is nebulized, rapidly evaporated in a heated region, [32, p. 56] and the resulting vapor passes a high-voltage needle that generates reagent ions. Solvents dominate the early ionization steps [22, p. 39] and effectively act as a reagent gas, transferring charge to neutral analytes mainly by proton transfer to form $[M + H]^+$.

Because APCI imparts relatively little internal energy, spectra typically show limited fragmentation and are well suited to quantitative LC-MS/MS, [32, p. 171] where signal is concentrated into a few precursor-ion peaks. A heated transfer tube and/or countercurrent curtain gas is used to decluster solvent-analyte adducts prior to ion extraction. [22, p. 39] APCI is often more efficient than ESI for less polar compounds, [1, p. 466] but its higher source temperatures can induce thermal degradation. [22, p. 39]

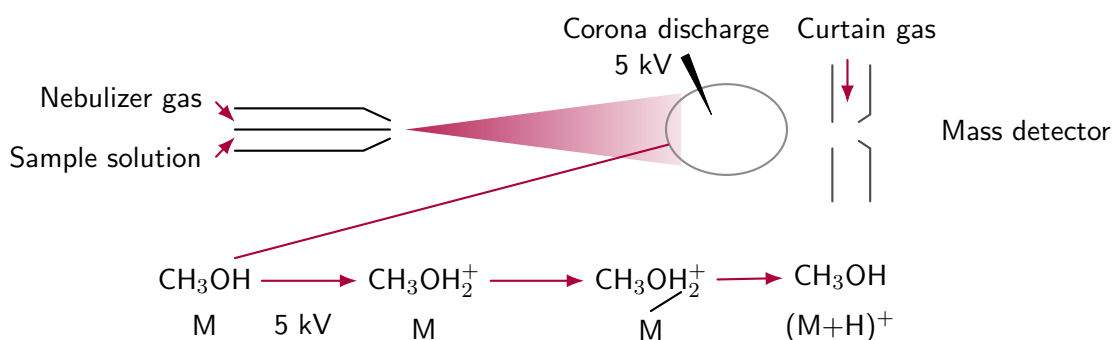


Figure 2.4: Schematic of the atmospheric pressure chemical ionization process

[22, p.39]

MALDI (Matrix-Assisted Laser Desorption/Ionization) [33] is a soft ionization technique [34]. that uses a laser to desorb and ionize analytes from a solid matrix. [32, p. 36] It is particularly effective for large biomolecules and polymers, [14, p. 33][32, p. 36] producing predominantly singly charged ions. [14, p. 35]. Tuning laser energy and matrix composition can optimize ionization efficiency and control fragmentation. A general schematic of the MALDI process is shown in Figure 2.5.

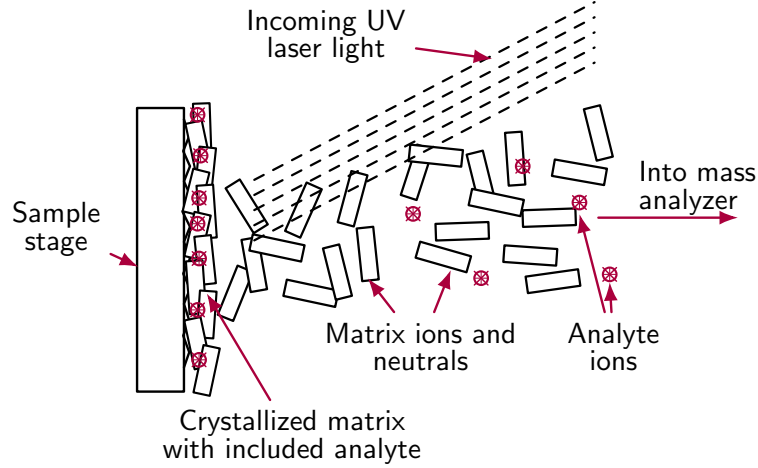


Figure 2.5: UV laser desorbs/ionizes co-crystallized matrix and analyte, creating an ion plume that is sent into the mass analyzer.

[22, p. 42]

2.2.2 Mass Analyzers

After ionization, the generated ions must be separated according to their mass-to-charge ratio m/z before detection. This separation is performed by the *mass analyzer*. [32, pp. 5-6] Different analyzer designs exploit electric and magnetic fields to distinguish ions with different m/z values. Among the most widely used analyzers are magnetic sector instruments, time-of-flight (TOF) analyzers, and quadrupole analyzers. [32, pp. 108-109]

Magnetic Sector Mass Analyzers

Magnetic sector analyzers separate ions according to their motion in a magnetic field. The motion of a charged particle with mass m and charge q in electric and magnetic fields is described by the Lorentz force equation

$$\left(\frac{m}{q}\right) \mathbf{a} = \mathbf{E} + \mathbf{v} \times \mathbf{B},$$

where $\mathbf{E}$ is the electric field, $\mathbf{B}$ the magnetic field, $\mathbf{v}$ the ion velocity, and $\mathbf{a}$ the acceleration.

Ions with identical mass-to-charge ratios experience identical forces and therefore follow identical trajectories inside the analyzer.

When ions enter a magnetic field perpendicular to their velocity, the Lorentz force bends their trajectory into a circular path. The balance between the magnetic force and the centripetal force yields

$$qvB = \frac{mv^2}{r},$$

with r being the radius of the circular path, which leads to

$$\frac{m}{q} = \frac{B^2 r^2}{2V},$$

assuming the ions were accelerated through a potential V . Ions with different m/z therefore follow trajectories with different radii and can be spatially separated. [35, pp. 1799-1800]

Time-of-Flight (TOF) Analyzers

Time-of-flight analyzers separate ions according to the time required to travel a fixed distance after acceleration. When ions are accelerated through a potential V , they acquire kinetic energy

$$\frac{1}{2}mv^2 = qV.$$

Solving for the velocity gives

$$v = \sqrt{\frac{2qV}{m}}.$$

In a field-free drift region of length L , the flight time is therefore

$$t = \frac{L}{v} = L\sqrt{\frac{m}{2qV}}.$$

Consequently, lighter ions reach the detector earlier than heavier ions, allowing determination of their m/z values. [36, p. 1152]

Quadrupole Mass Filter (QMF)

Quadrupole analyzers consist of four parallel rods to which a combination of radio-frequency (RF) and direct current (DC) voltages is applied. [37, p. 2] The resulting oscillating electric field produces ion trajectories that are stable only for ions within a specific range of mass-to-charge ratios. Ions with stable trajectories pass through the quadrupole to the detector, while ions with unstable trajectories collide with the rods and are removed. [38, p. 1] By scanning the applied

voltages, ions of increasing m/z can be transmitted sequentially, generating a mass spectrum. [37, p. 2]

Quadrupole Ion Trap (QIT)

The quadrupole ion trap operates on the same fundamental principle as the linear quadrupole analyzer described above, added with the ability of oscillating radiofrequency fields to selectively confine ions according to their mass-to-charge ratio. [39, p. 1156] The key distinction lies in the geometry of the electric field: whereas the linear quadrupole uses four parallel rods to create a two-dimensional confining potential that allows ions to pass through along the axial direction, the ion trap employs a three-dimensional quadrupolar field configuration that confines ions in all spatial dimensions within a central cavity. [40, p. 962] This fundamental geometric difference enables the QIT to accumulate and store ions over time [40, p. 973], rather than merely filtering them during transit, which has profound implications for sensitivity, scan modes, and the ability to perform multiple stages of mass analysis within a single device. [40, p. 984]

2.2.3 Detection

After ions have been separated by the mass analyzer, they must be detected and converted into a measurable electrical signal. [32, pp. 5-6] The detector is transforming the arrival of individual ions into quantifiable data that can be assembled into a mass spectrum. [32, p. 103]

The dominant detector in Electron Ionization Mass Spectrometry is the Electron Multiplier (EM). [32, p. 103]

An electron multiplier operates by converting a single incoming ion into a large cascade of electrons through the following sequence:

1. An ion hits the first dynode
2. Secondary electrons are emitted
3. These electrons are accelerated to the next dynode
4. The process repeats (10-20 stages)
5. Result: a measurable current pulse

This cascade amplification principle is illustrated in Figure 2.6.

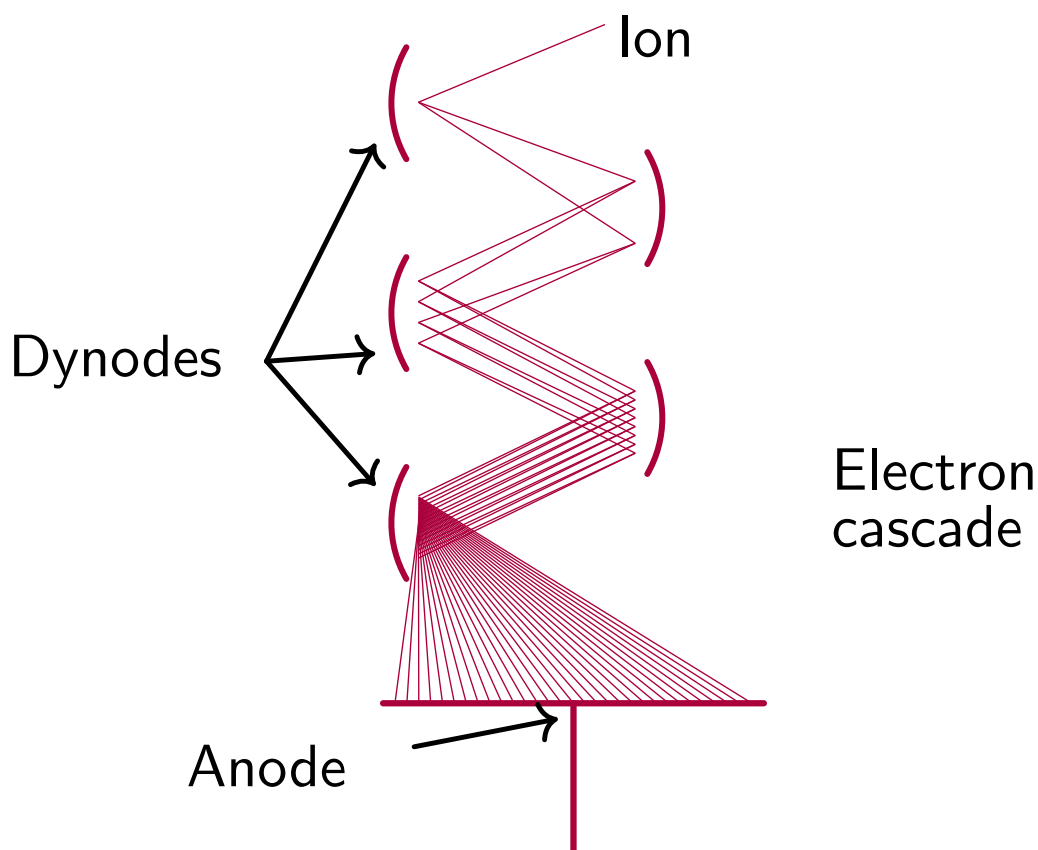


Figure 2.6: Schematic of an electron multiplier detector

[22, p. 50]

The amplification factor of an electron multiplier can reach 10^6 to 10^8 , enabling the detection of individual ions with high sensitivity. The resulting current pulses are digitized and recorded as a function of m/z , producing the final mass spectrum that serves as the basis for subsequent data analysis. [32, p. 104]

2.2.4 Manual Interpretation of the MS Spectrum

Data analysis transforms the raw spectrum into structural hypotheses: isotope patterns constrain elemental composition, [1, p. 67] while fragment ions and neutral-loss relationships constrain functional groups and connectivity. [26] Note, multiple candidate structures can be compatible with the same spectral evidence, particularly for isomers under hard ionization methods like EI. [41, p. 6996]

The following summarize a typical manual workflow [32, pp. 215-216] for interpreting EI mass spectra.

1. Determine the molecular ion peak (M) and its isotopic pattern

2. find isotope patterns (e.g., Cl, Br, S)

Table 2.1: sample isotope patterns

Element	Diagnostic pattern
Cl	3:1 doublet (M, M+2)
Br	1:1 doublet
S	small M+2 peak
C	^{13}C gives predictable M+1

3. determine the base peak (100%) which is the energetically favored fragmentation
4. analyze fragment differences and identify neutral losses
5. match fragmentation to chemical logic defined by reaction rules

These practical heuristics are grounded in specific EI reaction mechanisms, which are outlined in the following subsection.

2.2.5 Reactions in Electron Ionization

Fragmentation reactions in electron ionization mass spectrometry follow well-characterized patterns that are not random but are reproducible pathways determined by [24] the internal energy distribution and the intrinsic chemistry of the molecular ion. [42, p. 846] Understanding these reaction mechanisms is essential for interpreting EI spectra and for predicting fragmentation patterns from molecular structure. [42, p. 846]

They have been studied extensively by McLafferty and Tureček, see [11]. The following subsections describe some of the most important mechanisms.

Homolytic single-bond cleavage

The most straightforward fragmentation pathway is *simple bond cleavage*, in which a single bond in the molecular ion breaks homolytically, yielding a charged fragment and a neutral radical. The probability of cleavage at a particular bond depends primarily on bond strength: weaker bonds are more readily broken. [1, p. 30] For aliphatic hydrocarbons, C–C bonds adjacent to branching points or heteroatoms are particularly susceptible to cleavage. [1, p. 258]

Alpha Cleavage

Alpha cleavage is a characteristic fragmentation mode in molecules containing functional groups with heteroatoms. [32, pp. 220–221] In this process, the bond directly adjacent to the functional group is cleaved, producing a resonance-stabilized cation. [32, pp. 220–221] For instance, in ketones and aldehydes, cleavage of the C–C bond next to the carbonyl group generates an acylium ion RCO^+ , which is stabilized by the resonance structure $\text{R}-\text{C}\equiv\text{O}^+$. [1, pp. 234–235] This reaction pathway often produces intense peaks in the spectrum and serves as a diagnostic indicator for carbonyl-containing compounds. [1, pp. 234–235]

McLafferty Rearrangement

The *McLafferty rearrangement* is a hydrogen-transfer reaction characteristic of carbonyl compounds and other systems with abstractable γ -hydrogens.[32, p. 223] In the commonly depicted mechanism, a six-membered transition state facilitates the transfer of a hydrogen atom from the γ position to the carbonyl oxygen, accompanied by cleavage of the bond between the α and β carbons. [1, p. 265] The result is an enol radical cation and a neutral alkene. [32, p. 224] This rearrangement is particularly valuable for structural elucidation because it reveals connectivity information that would not be apparent from simple cleavage alone. [32, p. 224]

Charge-Induced Fragmentations

Beyond the radical-site-driven cleavages described above, fragmentation can also be initiated by the charge center itself. In such *charge-directed fragmentations*, the positive charge polarizes nearby bonds, facilitating heterolytic cleavage and producing even-electron cations and neutral radicals. [14, pp. 281]

2.3 Computational Mass Spectrometry

Computational tools significantly enhance MS capabilities for structure elucidation and compositional analysis. [15, pp. 1–2] Modern workflows integrate *in silico* fragmentation models, spectral database searches, and machine-learning methods to accelerate compound identification and enable high-throughput analysis. [24, p. 1] [15, p. 2]

In this section, we outline the key computational strategies in mass spectrometry, including library matching, *in silico* prediction, and *de novo* structure inference, before turning to their mathematical and algorithmic foundations.

2.3.1 Library Search

Library search is a spectrum-matching procedure used to identify an unknown compound by comparing its measured spectrum against a reference database of known spectra. [15, p. 8] Each entry in such a database typically stores a reference spectrum together with metadata such as molecular formula, structure, and acquisition conditions.

The comparison operates on two observable quantities: the m/z values of detected peaks and their relative intensities. A query spectrum is considered a match if a reference spectrum reproduces both the positions and the abundance pattern of its peaks sufficiently well. [22, p. 35]

A typical library search workflow proceeds as follows: [15, pp. 1, 5]

- (i) **Pre-processing:** the query spectrum is converted into a form suitable for comparison, for example by peak selection, intensity scaling, or related preprocessing steps.
- (ii) **Candidate filtering:** database entries may be restricted using precursor mass or precursor m/z information within a tolerance window, reducing the number of candidates that must be evaluated. [5, p. 14]
- (iii) **Scoring and ranking:** each candidate reference spectrum is assigned a similarity score against the query, and candidates are ranked accordingly. [15, p. 5]

The most widely used similarity metric is the *cosine similarity*, computed over the vector of peak intensities aligned by m/z : [43, p. 1]

$$\cos(\mathbf{q}, \mathbf{r}) = \frac{\mathbf{q} \cdot \mathbf{r}}{\|\mathbf{q}\| \|\mathbf{r}\|}, \quad (2.1)$$

where $\mathbf{q}$ and $\mathbf{r}$ are the intensity vectors of the query and reference spectrum, respectively. A score of 1 indicates a perfect match; lower values indicate increasing dissimilarity.

2.3.2 In Silico Fragmentation

In silico fragmentation predicts how a molecule may dissociate in a mass spectrometer and generates a theoretical spectrum from these simulated fragment ions. [7, p. 1] The standard workflow starts from a candidate molecular structure, applies computational fragmentation rules or learned models to produce plausible fragments, converts these fragments into a predicted spectrum, and finally compares the prediction against the measured spectrum using a similarity score. [7, p. 2] This approach is particularly useful when no exact match is available in spectral libraries, because it enables structure-aware candidate ranking beyond pure reference lookup. [15, pp. 1–2][24, p. 1] At the same time, prediction quality depends strongly on how well the

model captures real instrument conditions, ionization-dependent chemistry, and fragment intensities; consequently, performance can degrade for compounds or fragmentation pathways that are underrepresented in the training data or incompletely described by rule sets. [44, p. 1989]

When these *in silico* predictions remain ambiguous, the workflow may shift to *de novo* inference directly from spectral constraints.

2.3.3 De Novo Structure Prediction

De novo structure prediction denotes the inference of molecular structure from scratch, i.e., without relying on pre-existing templates or premade reference repositories [45, p. 1]. Importantly, *de novo* characterizes the *template-free* nature of the workflow and does not, by itself, imply a specific modeling paradigm (e.g., physics-based versus data-driven). Accordingly, *de novo* approaches may incorporate empirical heuristics, statistical learning, or mechanistic constraints, depending on the application. [46, pp. 701–702]

2.3.4 First Principles

In contrast, *first-principles* modeling emphasizes deriving predictions without empirical parameterization, i.e., without fitting to experimental data. [47, p. 1] In physics and chemistry, such first-principles treatments are often realized via *ab initio* electronic-structure methods.

Ab Initio MS Prediction

Ab initio MS prediction refers to computational modeling of mass spectra directly from molecular structure using nonempirical quantum-mechanical methods that are, in principle, independent of experimental fitting [48, p. 1921]. In this sense, *ab initio* approaches constitute first-principles modeling, because they aim to predict fragmentation behavior and resulting spectra from the underlying physics and chemistry of the molecule rather than from learned correlations. Operationally, *ab initio* MS prediction typically relies on quantum-chemical calculations (to obtain energetics and feasible fragmentation channels) combined with dynamical or statistical treatments to relate accessible pathways to predicted fragment distributions.

While *ab initio* methods can provide mechanistic insight and may achieve high accuracy for small molecules, their computational cost often limits scalability to large molecular libraries and complex systems. [46, p. 702]

2.4 Computational Representations of Molecules and Reactions

The encoding of molecular structures and reactions in a machine-processable form is essential for the feasibility of making algorithmic predictions. [49, p. 518]

In computational chemistry, molecules can be represented as graphs. [49, p. 521] Reactions are modeled as graph transformations that break and form bonds. [50, p. 18]

2.4.1 Molecular Graphs

To enable algorithmic processing of molecular structures, we represent them using an abstract graph model that separates topology (which atoms are connected) from optional geometric detail. [50, p. 18] This abstraction enables precise operations such as matching and rule-based transformation. [12, pp. 2–3] The formal definition used throughout this thesis is introduced next.

Formal definition

A molecular graph is typically defined as [51, p. 2] [52, p. 3]

$$G = (V, E) \tag{2.2}$$

where:

- V ... Vertices represent atoms
- E ... Edges represent chemical bonds between atoms

In Representations each node may carry attributes such as element type (e.g., C, N, O), formal charge, aromaticity, hybridization state, and isotope label. Likewise, each edge may encode bond order (single, double, or triple), aromatic character, and bond stereochemistry. [53, pp. 4–5] By construction, this basic molecular graph does not include geometric or electronic detail, such as 3D coordinates, molecular conformation, or wavefunction-level information; these require augmented graph models or separate geometric and electronic representations. [53, pp. 3–4] [54, p. 595]

2.4.2 Chemical Graph Transformation Systems

To model chemical reactions algorithmically, molecular graphs are combined with rewrite rules that encode local bond changes. [55, p. 76] In this view, a reaction is not an arbitrary edit of a graph, but a constrained transformation that preserves valid molecular structure while modifying a specific reaction center. [12, p. 2] [56, p. 6]

Double Pushout Graph Rewriting

A standard formalism is the double pushout (DPO) approach. [12, p. 3] A rule is given by

$$p = (L \xleftarrow{l} K \xrightarrow{r} R),$$

where L is the left-hand side pattern to be matched, R is the right-hand side pattern after rewriting, and K is the interface that is preserved. [55, p. 75] Given a match $m : L \rightarrow G$ in a host graph G , DPO rewriting constructs an intermediate graph D (deletion pushout complement) [57, p. 680] and then the product graph H (insertion pushout). Intuitively, parts in $L \setminus K$ are removed, parts in $R \setminus K$ are added, and K remains unchanged. [12, p. 3] [58, p. 4]

$$\begin{array}{ccccc} L & \xleftarrow{l} & K & \xrightarrow{r} & R \\ m \downarrow & & \downarrow & & \downarrow m^* \\ G & \xleftarrow{l^*} & D & \xrightarrow{r^*} & H \end{array}$$

In chemistry, this provides a precise representation of a reaction center: atoms and bonds in the preserved interface K define shared context, while changes between L and R encode bond breaking/forming events. [59, p. 5]

Derivation Graph

Building on the local rewrite step formalized above, derivation graphs – induced by repeated rule application – provide a global view that shows mechanistically plausible chemical states. [12, pp. 2–3]

A derivation graph, often also called a reaction network, is a higher-level graph that captures all transformation steps generated by a graph rewriting system. [60, Sec. 10] [56, p. 2] Whereas a molecular graph represents the internal structure of a single molecule, a derivation graph represents the *space of possible reactions* between many molecular states. [12, pp. 2–3]

Formally, each node corresponds to an object state (e.g., a molecule, multiset of molecules, or intermediate), and each directed edge corresponds to an application of a rewrite rule that transforms one state into another. [55, p. 75] If a rule p transforms state G into state H , this step is represented as

$$G \xrightarrow{p} H.$$

By collecting many such derivations, one obtains a directed graph whose topology encodes reachable states, alternative pathways, and cycles.

In computational chemistry, derivation graphs are useful for enumerating reaction pathways, identifying branching points, analyzing network connectivity, and ranking plausible mechanistic

routes under additional constraints (e.g., energy, rule priorities, or experimental evidence). They therefore provide a natural bridge between rule-based chemistry models and downstream tasks such as pathway search, and machine-learning-based scoring of candidate transformations. [56, p. 2]

2.5 Machine Learning in Mass Spectrometry

2.5.1 Problem Formulations

Two main problem formulations are common in the literature:

- **Structure-to-Spectrum Prediction:** Given a molecular structure, predict the corresponding mass spectrum. This is a forward problem that can be approached using mechanistic models, data-driven models, or hybrid approaches that combine both.
- **Spectrum-to-Structure Prediction:** Given a mass spectrum, predict the corresponding molecular structure. As this is an inverse problem, it is often ill-posed, with multiple candidate structures potentially consistent with the same spectrum.

Both tasks are important for different applications: structure-to-spectrum prediction is useful for in silico fragmentation and library generation, while spectrum-to-structure prediction is essential for de novo identification of unknown compounds.

Bidirectional models that can perform both tasks are also possible, and may leverage shared representations to improve performance on both tasks.

2.5.2 Case Study: Generating Molecular Fragmentation Graphs with Autoregressive Neural Networks

A prominent example of integrating physical fragmentation models with neural networks is the work of [8], which proposes an approach to predict tandem mass spectra by directly modeling fragmentation pathways as graphs. Rather than treating spectrum prediction as a black-box prediction task, this method learns to generate fragmentation graphs sequentially: at each step, the model predicts which bonds are most likely to break in the molecular ion, scores the resulting fragments, and iteratively builds a complete fragmentation landscape.

This hybrid approach is particularly relevant to computational MS because it bridges two historically separate paradigms: rule-based in silico fragmentation, which is interpretable but computationally expensive and relies on heuristic intensity prediction, and purely data-driven neural networks, which are fast but lack chemical interpretability. [8, p. 3420] By grounding

fragmentation prediction in explicit bond-breaking events, the model maintains mechanistic transparency while leveraging the expressiveness of neural networks for learning complex fragmentation preferences from data.

The method demonstrates several key strengths. Because predictions are generated as explicit fragmentation pathways, the model enables human-interpretable explanations: chemists can inspect why particular fragments appear and evaluate whether predicted breakages are chemically plausible. Also the approach shows particular utility for natural products and complex scaffolds where fragmentation patterns deviate from those captured by simple bond-breaking rules, improving metabolite identification from candidate databases. [8, p. 3427]

However, limitations remain. Prediction quality still depends on the distribution of training data; compounds or fragmentation pathways underrepresented in training tend to be mispredicted, and the model may struggle with instrument-specific quirks or unusual ionization regimes not seen during learning.

2.6 Interpretability in Chemical Machine Learning

2.6.1 Why Interpretability Matters

Interpretability in machine learning refers to the ability to understand and explain why a model makes a particular prediction. [61, p. 1]

For structure elucidation from MS data, an uninterpretable model that ranks candidate structures is of limited value if chemists cannot understand the reasoning. Conversely, models that expose their decision-making process [61, p. 2] —by highlighting which structural features drive spectral predictions or which fragmentation pathways are favored— enable hypothesis generation and accelerate scientific insight.

Furthermore, interpretability serves as a diagnostic tool: when model explanations reveal chemically implausible patterns (e.g., incorrect fragmentation rules or unphysical feature importance), this signals potential model failures or training-data biases that numeric performance metrics alone might not expose.

2.6.2 Classic and General Interpretability Methods

Saliency Maps and Gradient-Based Attribution

Saliency maps compute the gradient of the model output with respect to input features, visualizing regions of input space where small changes produce large changes in prediction. For molecular inputs, saliency can be backpropagated through graph neural networks to identify

atoms or bonds whose perturbation most affects the predicted spectrum. [62, p. 4] This approach is computationally efficient and provides local explanations for individual predictions. However, gradients can be noisy, especially near decision boundaries, and may not capture higher-order interactions or saturating effects. [63]

SHAP and Permutation-Based Methods

SHAP (SHapley Additive exPlanations) provides a principled, game-theoretic framework for feature attribution based on Shapley values from cooperative game theory. [64, p. 3] By computing the marginal contribution of each feature when iteratively included or excluded, SHAP assigns consistent importance scores that sum to the difference between the model prediction and a baseline. [64, p. 2] For molecular graphs, SHAP can be applied to atomic features or subgraph patterns, revealing which structural motifs drive predictions. Permutation-based importance methods offer similar insights by repeatedly shuffling features and measuring the resulting drop in model performance. [65, p. 23] These methods are model-agnostic and theoretically grounded, but computationally expensive for large molecules or high-dimensional representations.

Counterfactual Explanations

Counterfactual explanations identify minimal modifications to an input that would change the model output, answering the question: “What would need to change for a different prediction?” [66, p. 4] In the chemical domain, counterfactuals correspond to structural modifications—the loss or gain of functional groups, changes in substitution patterns, or alternative regioisomers—that would yield different predicted spectra. [67, pp. 2–3] These explanations are human-interpretable and actionable, but their discovery is computationally challenging, especially in high-dimensional chemical spaces with complex constraints (e.g., synthetic accessibility, valence rules).

2.6.3 Evaluation of Explanations

The quality of explanations must be evaluated in order to determine whether a method actually achieves its explainability objective and to enable comparison between alternative explanation methods. [68, p. 1] Zhou et al. describe explanation quality in terms of *interpretability* and *fidelity*. Interpretability concerns human understandability and includes the properties of *clarity*, *parsimony*, and *broadness*, whereas fidelity concerns how accurately the explanation reflects model behaviour and includes *completeness* and *soundness*. [68, p. 3]

Based on this view, explanation quality can be assessed through both human-centred and

functionality-grounded evaluation. Human-centred evaluation includes application-grounded and human-grounded studies and often considers subjective measures such as trust and confidence together with task-related performance. Functionality-grounded evaluation instead relies on formal proxy metrics tied to specific explainability properties. [68, p. 8] [68, p. 10]

Chapter 3

Data and Representation

3.1 Datasets

3.1.1 Public Databases

MassBank, pubChem, NIST

3.1.2 Custom Dataset Construction

3.2 Preprocessing - Generating the Synthetic Data

`mod` Derivation Graph reason, goal: gives info of parents of fragments

3.2.1 `mod` Strategy

Forward strategy builds a pipeline that

1. seeds the derivation graph with the molecule via `mod.addSubset`,
2. applies exactly one ionization step wrapped in `sub_group` to constrain it to the current derivation graph, and then
3. applies up to `frag_repeat` fragmentation steps,
4. but only keeps paths whose intermediate fragments satisfy both mass bounds and charge bounds via nested `amu_bound` and `charge_bound` filters.

Backward strategy

1. seeds the derivation graph with the provided fragment graphs via `mod.addSubset`,

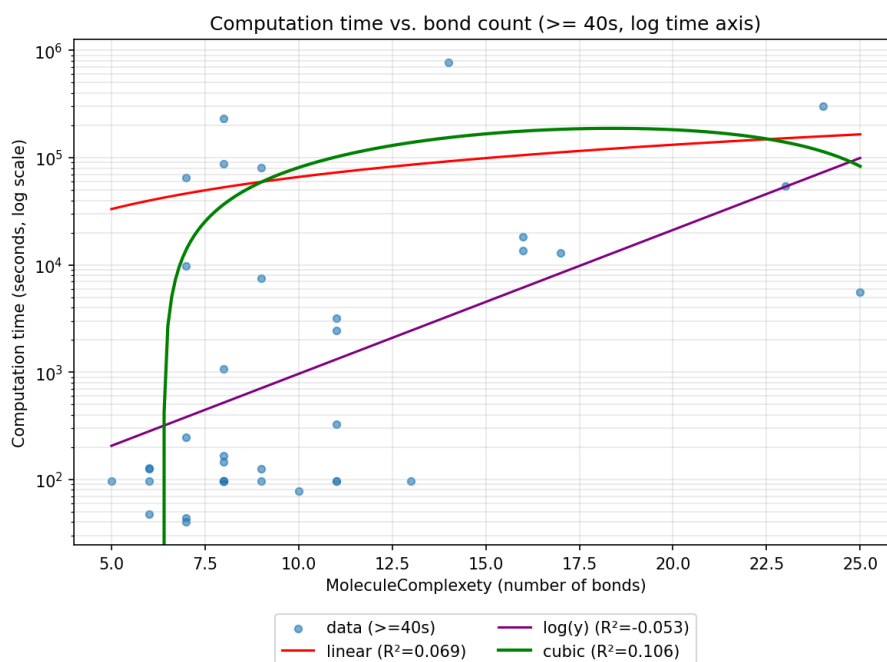


Figure 3.1: Regression of Computation time, filtered of molecules that took at least 40sec to compute

2. then applies up to `frag_repeat` fragmentation rules scoped to the derivation graph,
3. while filtering intermediates to keep only fragments within the given mass window and valid charge bounds (nested `amu_bound` inside `charge_bound`).

3.2.2 computation times

an improve is needed to make this aproach feasible also this is only a part of the rules that are known the compute time per molecule will rise even more

regression analasis, performed

assumption:

- Linearity
- Homoscedasticity
- Independence of Errors

threshold of 40sec as the computation might not be very good

3.3 Latend Representation Schemes

3.3.1 Molecular Graph Encoding

py-geometric

3.3.2 Fragment Graph Representation

technically the same as the Molecular Graph Vertex is a Atom with attributes like element type

Edge is a Bond with attributes bond strengh half bond strength is aromaticity

3.3.3 Spectral Vector and Embedding Representation

latend represantation of the ms spectra

3.3.4 Fragment–Spectrum Mapping

How spectra and fragments are aligned. The “fragment graph” concept - linking structure and spectral space.

Chapter 4

Model Architecture

4.1 Overview of the Learning Framework

4.2 Forward Model: Structure-to-Spectrum Prediction

4.2.1 GNN-Based Encoder

4.2.2 Fragment Predictor or Spectral Generator

4.3 Backward Model: Spectrum-to-Structure Reconstruction

4.3.1 Reinforcement Learning Environment

4.3.2 Reward Design and Shaping

4.4 Bidirectionllity

self supervised learning

4.5 Integration of MØD Hypergraph Derivations

Chapter 5

Experiments and Evaluation

5.1 Experimental Setup

5.1.1 Data Splits and Training Protocols

5.1.2 Evaluation Metrics

cosine jaccard index tanimoto edit-distances

wasserstein metrics

5.2 Forward Model Evaluation

5.2.1 Spectral Similarity and Fragment Prediction

5.3 Backward Model Evaluation

5.3.1 Structure Reconstruction and Validity

5.3.2 Reward Convergence and Exploration Efficiency

5.4 Ablation Studies

without fragment parent awareness, or in other words not using the derivation graph

5.5 Comparison with Baseline Models

CFM-ID,

MSNovelist (<https://doi.org/10.1038/s41592-022-01486-3>),

MassGenie (<https://doi.org/10.3390/biom11121793>) ...

Chapter 6

Discussion

6.1 Benefits of Fragment-Aware Modeling

How fragments improve interpretability.

6.2 Challenges

ambiguity, combinatorial explosion

6.3 potential improvements & future work

speed up graph transformation, especially predicates use one class of molecules

6.4 Summary of Contributions

Manuel Uhler

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Appendix A

My appendix

A.1 MØD Rule Sets and Strategy Definitions

A.2 Hyperparameter Tables and Configurations

A.3 Github Repository

<https://github.com/phireiser/mass-spec>